



DBMS 04 | Normal Form | CS & IT | GATE 2024 FastTrack Batch

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Summary

AI Notes

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Text Notes

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📷 28

[MCQ]

[2 Marks]

#Q. Consider the relation R (P, Q, S, T, X, Y, Z, W) with the following functional dependencies.
 $PQ \rightarrow X; P \rightarrow YX; Q \rightarrow Y; Y \rightarrow ZW$
Consider the decomposition of the relation R into the constituent relations according to the following two decomposition schemes.
 $D_1: R = [(P, Q, S, T); (P, T, X); (Q, Y); (Y, Z, W)]$
 $D_2: R = [(P, Q, S); (T, X); (Q, Y); (Y, Z, W)]$
Which one of the following options is correct?

A

D₁ is a lossless decomposition, but D₂ is a lossy decomposition.

B

D₁ is a lossy decomposition, but D₂ is a lossless decomposition.

C

Both D₁ and D₂ are lossless decomposition.

D

Both D₁ and D₂ are lossy decomposition.

▶ 4:25

$R(PQSTXYZW) [PQ \rightarrow X, P \rightarrow YX, Q \rightarrow Y, Y \rightarrow ZW]$

$D_1: [(PQST)^{R_1} (PTX)^{R_2} (QY)^{R_3} (YZW)^{R_4}]$

$D_2: [(PQS)^{R_1} (TX)^{R_2} (QY)^{R_3} (YZW)^{R_4}]$

$D_1: R_1(PQST) \cap R_2(PTX) = PT$
 $(PT)^+ = [PTYXZW]$ Super key of R_2 ①, is Lossless

$R_1(PQSTX) \cap R_3(QY) = Q$
 $(Q)^+ = [QYZW\dots]$ Super key of R_3

$R_{123}(PQSTXY) \cap R_4(YZW) \Rightarrow Y \Rightarrow (Y)^+ = [YZW]$ sk of R_4

▶ 8:49

$R(PQSTXYZW) [PQ \rightarrow X, P \rightarrow YX, Q \rightarrow Y, Y \rightarrow ZW]$

$D_1: [(PQST)^{R_1} (PTX)^{R_2} (QY)^{R_3} (YZW)^{R_4}]$

$D_2: [(PQS)^{R_1} (TX)^{R_2} (QY)^{R_3} (YZW)^{R_4}]$

$R_1(PQS) \cap R_3(QY) = (Q)^+ = [QYZW\dots]$ Super key of R_3

$R_{13}(PQSY) \cap R_4(YZW) \Rightarrow Y \Rightarrow (Y)^+ = [YZW]$ sk of R_4

$R_{134}(PQSYZW) \cap R_2(TX)$

Lossy

▶ 11:53



Topic : Normal Forms



First Normal Form [1NF]:

A Relation Schema R is in 1NF iff R does not contain any Multivalued Attribute.

A Relation R is in 1NF iff ^{OR} all attribute of R are atomic.

STUDENT

Sid	Sname	Course
S ₁	A	C/JAVA

Not in 1NF

Multivalued Attribute

STUDENT

Sid	Sname	Course
S ₁	A	C
S ₁	A	JAVA

R is in 1NF.

▶ 27:35



Topic : Normal Forms



Note Default RDBMS is in 1NF.

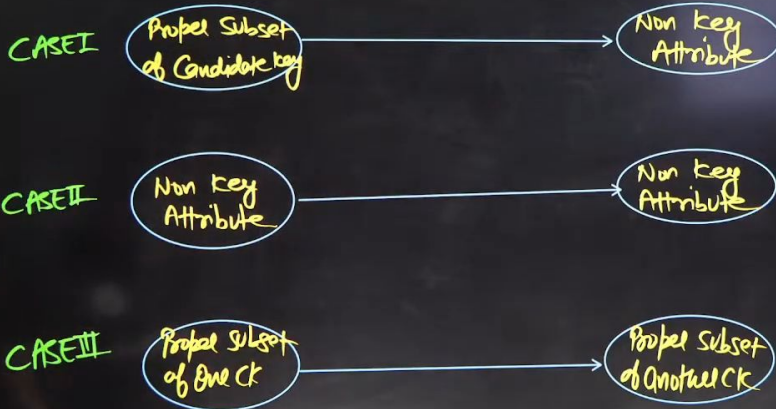
Note RDBMS does not accept any Multivalued Attribute.

Note 1NF Ensured by Candidate Key.

Redundancy Level: $1NF > 2NF > 3NF > BCNF$

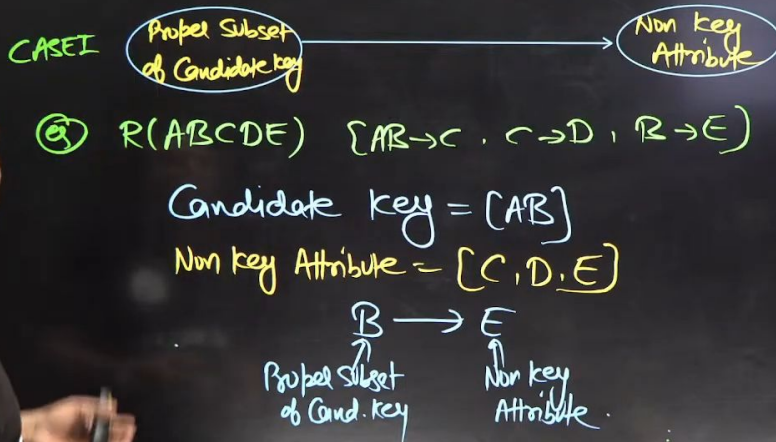
▶ 30:01

Possible Non Trivial FD Which cause Redundancy 18:34



▶ 34:09

Possible Non Trivial FD Which cause Redundancy 18:37



▶ 36:47

Possible Non Trivial FD Which cause Redundancy 18:41

CASE II

Non key Attribute

Non key Attribute

③ R(ABCD) [AB → C, C → D]

Candidate key = [AB]

Non key Attribute = [C, D]

C → D
Non key Attribute Non key Attribute

④ R(ABC) [A → B, B → C]

Candidate key = [A]

Non key Attribute = [B, C]

B → C
Non key Attribute Non key Attribute

▶ 40:58

Possible Non Trivial FD Which cause Redundancy 18:44

CASE III

Proper Subset of One CK

Proper Subset of another CK

⑤ R(ABCD) [AB → CD, D → A]

Candidate key = [AB, DB]

D → A
Proper Subset of One CK Proper Subset of another CK

▶ 44:29

Topic : Normal Forms

Partial FD

$AB \rightarrow C$ is a Partial FD

ibb

$$\begin{array}{l} A \rightarrow C \\ \textcircled{B} \\ B \rightarrow C \end{array}$$

▶ 1:03:20

Topic : Normal Forms

Partial FD

$AB \rightarrow C$ is a Partial FD

ibb

$$\begin{array}{l} A \rightarrow C \\ \textcircled{B} \\ B \rightarrow C \end{array}$$

$A \in X$

$X \rightarrow y$ is Partial FD.

$(X - A) \rightarrow y$

$\textcircled{B} \quad \overset{X}{AB} \rightarrow \overset{y}{C}$ is P.F.D

Remove 'A' $\left\{ \begin{array}{l} B \rightarrow C \\ \textcircled{B} \end{array} \right.$

Remove B $\left\{ \begin{array}{l} B \rightarrow C \\ A \rightarrow C \end{array} \right.$

▶ 1:05:25

Normal Forms



Navathe Book.

15.3.5 Second Normal Form

Second normal form (2NF) is based on the concept of full functional dependency. A functional dependency $X \rightarrow Y$ is a **full functional dependency** if removal of any attribute A from X means that the dependency does not hold any more; that is, for any attribute $A \in X$, $(X - \{A\})$ does *not* functionally determine Y . A functional dependency $X \rightarrow Y$ is a **partial dependency** if some attribute $A \in X$ can be removed from X and the dependency still holds; that is, for some $A \in X$, $(X - \{A\}) \rightarrow Y$. In Figure 15.3(b), $\{Ssn, Pnumber\} \rightarrow Hours$ is a full dependency (neither $Ssn \rightarrow Hours$ nor $Pnumber \rightarrow Hours$ holds). However, the dependency $\{Ssn, Pnumber\} \rightarrow Ename$ is partial because $Ssn \rightarrow Ename$ holds.

Definition. A relation schema R is in 2NF if every nonprime attribute A in R is fully functionally dependent on the primary key of R .

$AB \rightarrow C$ is Full FD

ibb Neither $A \rightarrow C$

nor $B \rightarrow C$

▶ 1:07:35

Topic : Normal Forms



Korth Book

- ❑ A functional dependency $\alpha \rightarrow \beta$ is called a partial dependency if there is a proper subset γ of α such that $\gamma \rightarrow \beta$. We say that β is partially dependent on α . A relation schema R is in second normal form (2NF) if each attribute in R meets one of the following criteria:
- ❑ It appears in candidate key.
- ❑ It is not partially dependent on a candidate key.

▶ 1:17:29



Topic : Normal Forms



Identify the P.F.D ?

$R(ABCDEF) \{ABC \rightarrow DE, DE \rightarrow ABC, AB \rightarrow D, DE \rightarrow F, E \rightarrow C\}$

- (i) $ABC \rightarrow D$
- (ii) $AE \rightarrow C$
- (iii) $AF \rightarrow D$
- (iv) $AB \rightarrow D$
- (v) $AC \rightarrow D$
- (vi) $BC \rightarrow D$
- (vii) $DE \rightarrow C$
- (viii) $AB \rightarrow F$

▶ 1:20:28



Topic : Normal Forms



Second Normal Form [2NF]: A Relation R is in 2NF if every Non Prime Attribute is fully Dependent on the key.

(OR)

A Relation R is in 2NF iff

- ① R is in 1NF
- ② R does not contain this type of FD.

Proper Subset
of Candidate key

Violation of
2NF.

Non key
Attribute.

▶ 1:38:10

Emp Proj (Ssn, Pnumber, Hours, Ename, Pname, Plocation)

Candidate key = {Ssn, Pnumber} FDI : Ssn Pnumber $\rightarrow$ Hours.

Non key / Non Prime Attribute = {Hours, Ename, Pname, Plocation} FDII : Ssn $\rightarrow$ Ename

FDIII : Pnumber $\rightarrow$ Pname Plocation.

$(\text{Ssn Pnumber})^+ = \{ \text{Ssn Pnumber Hours Ename Pname Plocation} \}$

Ssn Pnumber $\rightarrow$ Ename

Ssn $\rightarrow$ Ename

Ssn Pnumber $\rightarrow$ Ename is Partial FD.

Ssn Pnumber $\rightarrow$ Pname is Partial FD.
Pnumber $\rightarrow$ Pname

▶ 1:50:24

Topic : Normal Forms

Second normal form

Defination :

Definition: A relation schema R is in 2NF if every nonprime attribute A in R is fully functionally dependent on the primary key of R.

EMP_PROJ

Ssn	Pnumber	Hours	Ename	Pname	Plocation
-----	---------	-------	-------	-------	-----------

FD1: Ssn $\rightarrow$ Hours

FD2: Ssn $\rightarrow$ Ename

FD3: Pnumber $\rightarrow$ Pname Plocation

Not in 2NF

2NF Normalization

EP1

Ssn	Pnumber	Hours
-----	---------	-------

FD1: Ssn $\rightarrow$ Hours

EP2

Ssn	Ename
-----	-------

FD2: Ssn $\rightarrow$ Ename

EP3

Pnumber	Pname	Plocation
---------	-------	-----------

FD3: Pnumber $\rightarrow$ Pname Plocation

▶ 1:52:34

(eg) $R(ABCDE) \{AB \rightarrow C, C \rightarrow D, B \rightarrow E\}$

Candidate key = $[AB]$

Non key Attribute = $[C, D, E]$

AB is C.K that means

$AB \rightarrow E$

$B \rightarrow E$

So $AB \rightarrow E$ is Partial FD.
Here Non Prime Attribute 'E' is Partially
Dependant on C.K AB .

$B \rightarrow E$
↑
Proper subset
of CK
Non key
Attribute

Not in 2NF

▶ 1:59:53

[MCQ]

#Q. Let $R(A, B, C, D, E, P, G)$ be a relational schema in which the following functional dependencies are known to hold:

$AB \rightarrow CD, DE \rightarrow P, C \rightarrow E, P \rightarrow C$ and $B \rightarrow G$

The relational schema R is

(GATE: 2 Marks)

A

In BCNF

B

In 3NF, but not in BCNF

In 2NF, but not in 3NF

Not in 2NF

Proper subset
of CK → Non key
Attribute
Not in 2NF

▶ 2:01:37



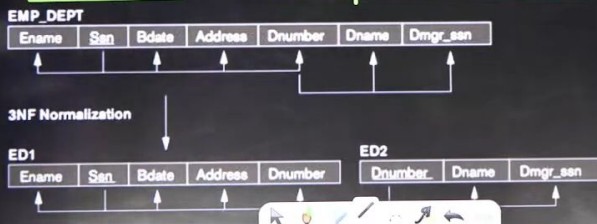
Topic : Normal Forms



Third Normal Form

Definition: According to Codd's original definition, a relation schema R is in 3NF if it satisfies 2NF and no nonprime attribute of R is transitively dependent on the primary key. OR

Definition: A relation schema R is in third normal form (3NF) if, whenever a nontrivial functional dependency $X \rightarrow A$ holds in R either (a) X is a superkey of R, or (b) A is a prime attribute of R.



▶ 2:04:56

Third Normal Form [3NF] :

$R(ABC) \{A \rightarrow B, B \rightarrow C\}$

Candidate key = [A]

Non key Attribute = [B, C]

Definition I

$A \rightarrow B, B \rightarrow C$

$A \rightarrow C$ is transitive FD

Non Prime Attribute 'C' is transitively
dependent on Candidate (Primary) key.
 So Not in 3NF.

Definition II

$A \rightarrow B$ 3NF ✓

$B \rightarrow C$ 3NF ✗

So Not in 3NF

$X \rightarrow Y$
 either X: Super key
OR
 Y: key/Prime Attribute

A is Super key.

B is Not Super key

(a)
 C is Non Prime Attribute

▶ 2:13:49



Topic : Boyce – Codd Normal Form



Definition: A relation schema R is in BCNF if whenever a nontrivial functional dependency $X \rightarrow A$ holds in R, then X is a superkey of R.

$X \rightarrow y$
 X : Super key.

▶ 2:15:13

[MCQ]



#Q. In a relational data model, which one of the following statements is TRUE?

- ☐ A A relation with only two attributes is always in BCNF.
- ☐ B If all attributes of a relation are prime attributes, then the relation is in BCNF.
- ☐ C Every relation has at least one non-prime attribute.
- ☐ D BCNF decompositions preserve functional dependencies.

▶ 2:15:17

[MCQ]



#Q. Let $R(A, B, C, D, E, P, G)$ be a relational schema in which the following functional dependencies are known to hold:
 $AB \rightarrow CD$, $DE \rightarrow P$, $C \rightarrow E$, $P \rightarrow C$ and $B \rightarrow G$.
The relational schema R is

- ☐ A In BCNF
- ☐ B In 3NF, but not in BCNF
- ☐ C In 2NF, but not in 3NF
- ☐ D Not in 2NF

▶ 2:15:40

[MSQ]



#Q. Consider the following statements:
 S_1 : If every attribute is prime attribute in R , then Relation R will always be in BCNF.
 S_2 : Any Relation with two Attribute is in 3 NF and 2 NF.
 S_3 : If every key of relation R is a simple candidate key (No composite key) then the relation R not always in NF.
 S_4 : In BCNF there is always a lossless join and Dependency Preserving Decomposition.

Which of the above statement are incorrect

- | | |
|----------------------------------|----------------------------------|
| ☐ A S_1 | ☐ B S_2 |
| ☐ C S_3 | ☐ D S_4 |

▶ 2:15:43

[MCQ]

[2 Marks]



#Q. The relation scheme student Performance (name, courseNO, rollNo, grade) has the following functional dependencies:

name, courseNo $\rightarrow$ grade

RollNo, courseNo $\rightarrow$ grade

name $\rightarrow$ rollNo

rollNO $\rightarrow$ name

The highest normal form of this relation scheme is

A 2 NF

B 3 NF

C BCNF

D 4 NF

▶ 2:15:46

[MCQ]

[1 Marks]



#Q. In a relational data model, which one of the following statements is TRUE?

Relation with only two attributes is always in BCNF.

If all attributes of a relation are prime attributes, then the relation is in BCNF.

Every relation has at least one non-prime attribute.

BCNF decompositions preserve functional dependencies.

▶ 2:15:48

[NAT]

[1 Marks]



#Q. Consider a relation $R(A, B, C, D, E)$ with the following three functional dependencies.

$AB \rightarrow C$; $BC \rightarrow D$; $C \rightarrow E$;

The number of super keys in the relation R is _____.

▶ 2:15:50

[MCQ]

[2 Marks]



#Q. Consider a relational table R that is in 3 NF, but not in BCNF. Which one of the following statements is TRUE?

R has a non-trivial functional dependency $X \rightarrow A$, where X is not a superkey and A is a prime attribute.

R has a non-trivial functional dependency $X \rightarrow A$, where X is not a superkey and A is a non-prime attribute and X is not a proper subset of any key.

R has a non-trivial functional dependency $X \rightarrow A$, where X is not a superkey and A is a non-prime attribute and X is a proper subset of some key.

A cell in R holds a set instead of an atomic value.

▶ 2:15:53

[MCQ]

[2 Marks]



#Q. Let the set of functional dependencies $F = \{QR \rightarrow S, R \rightarrow P, S \rightarrow Q\}$ hold on a relation schema $X = (PQRS)$. X is not in BCNF. Suppose X is decomposed into two schemas Y and Z , where $Y = (P R)$ and $Z = (Q R S)$.

Consider the two statements given below.

I. Both Y and Z are in BCNF

II. Decomposition of X into Y and Z is dependency preserving and lossless

Which of the above statements is/are correct?

A

Both I and II

B

I only

C

II only

D

Neither I nor II

▶ 2:15:57